



Generate a formal lecture scribe for the provided lecture of the course CSE 400: Fundamentals of Probability in Computing. The output must be a strictly exam-oriented academic reference document, suitable for reading-based / closed-book revision. A student should be able to fully revise the lecture using only this scribe, without consulting any external material. Context Restriction (Strictly Enforced): Use only the provided lecture PowerPoint slides. Do not use external knowledge, textbook material not explicitly shown in the slides, inferred intuition, reinterpretations, or added explanations. Every definition, formula, example, derivation, and statement must be directly traceable to the provided slides. Required Content (Include only if present in the lecture): Formal definitions and notation Explicit assumptions and stated conditions Clearly stated theorems, propositions, or named results Proofs or proof sketches written step by step, exactly as presented in the lecture Worked examples with all intermediate steps shown Structural Expectations (Mandatory): Follow the exact topic order and logical flow of the lecture Organize the scribe using clear LaTeX sectioning (`\section`, `\subsection`) Present definitions, theorems, propositions, and formulas in clearly identifiable mathematical environments Clearly label Proof and Solution blocks so they are easy to locate during exam revision Exclude any administrative, instructional, or non-examinable content Strict Exclusions: Do not add: New examples or problems Intuition, motivation, or explanatory commentary beyond what appears in the lecture Simplifications, summaries, or textbook-style elaborations Any reasoning or explanation not explicitly present in the lecture slides Output Format (Non-Negotiable): Output only complete LaTeX source code, enclosed within a single `\documentclass ... \end{document}` block Do not include any plain text, commentary, or explanations outside the LaTeX code The LaTeX document must explicitly include: Course name Lecture number (as indicated in the provided slides) Group No.: S1 G17 Date The final output must be a faithful, well-structured, exam-ready reconstruction of the

`</> LaTeX`

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\documentclass[12pt]{article}
\usepackage{amsmath, amssymb}

\begin{document}

\title{CSE 400: Fundamentals of Probability in Computing\\
Lecture 20: Linear Transformations and Expectations of Multiple RVs}
\author{Group No.: S1 G17}
\date{March 19, 2025}
\maketitle

\section{Random Vector Representation}

\subsection{Vector Form of Multiple Random Variables}

For random variables  $X_1, X_2, \ldots, X_N$ , we represent them as a vector:
\begin{matrix}
\left[ \right. \\
X = \\
\begin{bmatrix}
X_1 \\
X_2 \\
\vdots \\
X_N
\end{bmatrix} \\
\end{matrix}

\textbf{Definition:}  $X$  is called a random vector.

\section{Mean Vector}

\subsection{Expectation of a Random Vector}

\textbf{Definition:} The mean vector of  $X$  is:
\begin{matrix}
\left[ \right. \\
\mu_X = E[X] \\
\end{matrix}

Example structure:
\begin{matrix}
\left[ \right. \\
\mu_X = \\
\begin{bmatrix}
E[X_1] \\
E[X_2] \\
\vdots \\
E[X_N]
\end{bmatrix} \\
\end{matrix}
```

`\section{Correlation Matrix}`

`\subsection{Correlation Between Multiple RVs}`

`\textbf{Definition:}`

$$\mathbf{R}_{XX} = E[XX^T]$$

Element-wise:

$$R_{XX}(i,j) = E[X_i X_j], \quad i,j \leq N$$

Matrix form:

$$\mathbf{R}_{XX} = \begin{bmatrix} E[X_1X_1] & E[X_1X_2] & \cdots & E[X_1X_N] \\ E[X_2X_1] & E[X_2X_2] & \cdots & E[X_2X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_NX_1] & E[X_NX_2] & \cdots & E[X_NX_N] \end{bmatrix}$$

`\textbf{Property:}` $\mathbf{R}_{XX}$ is symmetric.

`\section{Covariance Matrix}`

`\subsection{Definition}`

$$\mathbf{C}_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

Matrix structure:

$$\mathbf{C}_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

`\textbf{Property:}`

$$\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$$

`\section{Linear Transformation of Random Vectors}`

`\subsection{Linear Model}`

$$\begin{bmatrix} Y \\ \vdots \end{bmatrix} = A \begin{bmatrix} X \\ \vdots \end{bmatrix} + b$$

For N -dimensional vectors:

$$Y_i = \sum_{j=1}^N a_{ij} X_j + b_i$$

Mean of Transformed Vector

Definition

$$\mu_Y = E[Y]$$

Derivation

$$\begin{aligned} \mu_Y &= E[AX + b] \\ &= E[AX] + E[b] \end{aligned}$$

Result:

$$\mu_Y = A\mu_X + b$$

Correlation Matrix Transformation

Definition

$$R_{YY} = E[YY^T] = E[(AX + b)(AX + b)^T]$$

Expansion

$$R_{YY} = E[AXX^T A^T + AXb^T + bX^T A^T + bb^T]$$

Applying Expectation

$$R_{YY} = A E[XX^T] A^T + A E[X] b^T + b E[X]^T A^T + bb^T$$

Final Result:

$$\begin{aligned} & \\ & \end{aligned}$$

$$R_{\{YY\}} = A R_{\{XX\}} A^T + A \mu_X b^T + b \mu_X^T A^T + b b^T$$

Covariance Matrix Transformation

Key Observation

$$Y - \mu_Y = (AX + b) - (A\mu_X + b) = A(X - \mu_X)$$

Derivation

$$C_{\{YY\}} = E[(Y - \mu_Y)(Y - \mu_Y)^T]$$

$$= E[A(X - \mu_X)(X - \mu_X)^T A^T]$$

Simplified Result:

$$C_{\{YY\}} = A C_{\{XX\}} A^T$$

Note: Shift b does not affect covariance.

Summary

If $Y = AX + b$:

$$\mu_Y = A\mu_X + b$$

$$C_{\{YY\}} = A C_{\{XX\}} A^T$$

$$R_{\{YY\}} = A R_{\{XX\}} A^T + A \mu_X b^T + b \mu_X^T A^T + b b^T$$

Example: Numerical Linear Transformation

Given:

$$\mu_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

```
\end{bmatrix}
\]
```

```
\[
R_{XX} =
\begin{bmatrix}
13 & 2 & 3 \\
2 & 10 & 3 \\
3 & 3 & 10
\end{bmatrix}
\]
```

\subsection{(a) Mean Vector}

```
\[
\mu_Y = A\mu_X
\]
```

```
\[
=
\begin{bmatrix}
1 & 0 & -1 \\
0 & -1 & 1 \\
-1 & 1 & 0
\end{bmatrix}
\begin{bmatrix}
2 \\
-1 \\
1
\end{bmatrix}
=
\begin{bmatrix}
1 \\
2 \\
-3
\end{bmatrix}
\]
```

\subsection{(b) Correlation Matrix}

Step 1:

```
\[
A R_{XX} =
\begin{bmatrix}
10 & -1 & -7 \\
1 & -7 & 7 \\
-11 & 8 & 0
\end{bmatrix}
\]
```

Step 2:

```
\[
R_{YY} = (A R_{XX}) A^T =
\begin{bmatrix}
17 & -6 & -11 \\
-6 & 14 & -8 \\
-11 & -8 & 19
\end{bmatrix}
\]
```

\]

\subsection{(c) Covariance Matrix}

\[

$$C_{\{YY\}} = R_{\{YY\}} - \mu_Y \mu_Y^T$$

\]

\[

$$\mu_Y \mu_Y^T =$$

\begin{bmatrix}

$$1 \quad 2 \quad -3$$

\end{bmatrix}

\begin{bmatrix}

$$1 \quad 2 \quad -3$$

\end{bmatrix}

=

\begin{bmatrix}

$$1 \quad 2 \quad -3 \quad \backslash \backslash$$

$$2 \quad 4 \quad -6 \quad \backslash \backslash$$

$$-3 \quad -6 \quad 9$$

\end{bmatrix}

\]

\[

$$C_{\{YY\}} =$$

\begin{bmatrix}

$$17 \quad -6 \quad -9 \quad \backslash \backslash$$

$$-6 \quad 14 \quad -8 \quad \backslash \backslash$$

$$-9 \quad -8 \quad 19$$

\end{bmatrix}

-

\begin{bmatrix}

$$1 \quad 2 \quad -3 \quad \backslash \backslash$$

$$2 \quad 4 \quad -6 \quad \backslash \backslash$$

$$-3 \quad -6 \quad 9$$

\end{bmatrix}

=

\begin{bmatrix}

$$16 \quad -8 \quad -8 \quad \backslash \backslash$$

$$-8 \quad 10 \quad -2 \quad \backslash \backslash$$

$$-8 \quad -2 \quad 10$$

\end{bmatrix}

\]

\section{Gaussian Distributions}

\subsection{Univariate Gaussian (1D)}

\[

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

\]

\subsection{Joint Gaussian (2D)}

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} \exp\left(-\frac{1}{2(1-\rho_{XY}^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right)$$

`\section{Multivariate Gaussian Distribution}`

$$X = [X_1, X_2, \ldots, X_N]^T$$

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)\right)$$

`\section{Example: 2D Joint Gaussian}`

`\subsection{Determinant}`

$$\det(C_{XX}) = \sigma_1^2 \sigma_2^2 (1-\rho^2)$$

`\subsection{Inverse}`

$$C_{XX}^{-1} = \frac{1}{\det(C_{XX})} \begin{bmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{bmatrix}$$

`\subsection{Quadratic Form}`

$$Q(x) = (x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)$$

$$Q(x) =$$


```

\frac{1}{1-\rho^2}
\left[
\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2
-2\rho \left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right)
+\left(\frac{x_2-\mu_2}{\sigma_2}\right)^2
\right]
\]

```

```

\section{Uncorrelated Gaussian Variables}

```

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\[
f_X(x) = \prod_{n=1}^N
\frac{1}{\sqrt{2\pi}\sigma_n}
\exp\left(-\frac{(x_n-\mu_n)^2}{2\sigma_n^2}\right)
\]

```

```

\textbf{Condition:}

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\[
\mathrm{Cov}(X_i,X_j)=0, \quad i\neq j
\]

```

```

\section{Homework}

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\[
\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)
\]

```

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\end{document}

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